

1. Administrative & Study Identification

Full Study Title: A Phase 1, Open-label Study of Anti-CEACAM5 Antibody-Drug Conjugate Precemtabart Tocentecan (M9140) in Chinese Participants With Solid Tumors **Short Title/Acronym:** PROCEADE-CRC-02 **Protocol Number:** PROCEADE-CRC-02 **NCT Number:** NCT06806046 **Sponsor Name:** Merck KGaA, Darmstadt, Germany (Merck Healthcare KGaA) **Investigational Product:** Precemtabart tocentecan (M9140 / Precem-TcT), an Anti-CEACAM5 Antibody-Drug Conjugate (ADC) **Phase of Development:** Phase 1 **Medical Monitor:** Merck KGaA Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective **Primary Objective:** To evaluate the safety, tolerability, and early clinical activity of M9140 in Chinese participants with locally advanced or metastatic colorectal cancer (CRC). **Secondary Objectives:** To characterize the pharmacokinetics (PK) of M9140 and document its preliminary antitumor efficacy (e.g., Objective Response Rate, Progression-Free Survival).

2.2 Background & Rationale Carcinoembryonic antigen-related cell adhesion molecule 5 (CEACAM5) is a cell surface glycoprotein that is highly overexpressed on tumor cells (including colorectal cancer) and associated with tumor differentiation, invasion, and metastasis, but has low expression in normal tissues. Precemtabart tocentecan (M9140) is an investigational ADC designed to deliver a cytotoxic topoisomerase 1 (TOP1) inhibitor payload directly to CEACAM5-expressing tumor cells. The payload disrupts DNA replication, driving targeted cell death as well as a bystander effect to neighboring tumor cells. This Phase 1 trial assesses the specific safety profile, dose tolerance, and early clinical activity of M9140 in the Chinese population.

3. Study Design & Methodology

3.1 Design Framework **Study Type:** Interventional **Control Type:** Single-arm **Blinding:** Open-label **Randomization:** N/A

3.2 Planned Enrollment & Duration **Target \$N\$:** Approximately 12 participants **Number of Sites:** Multiple clinical sites in China **Estimated Study Start:** 20-Dec-2024 **Estimated Primary Completion:** 07-May-2026

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Participants with documented histopathological diagnosis of locally advanced or metastatic CRC who were intolerant/refractory to, or progressed after, standard systemic therapies (must include a fluoropyrimidine, irinotecan, a platinum agent, an anti-EGFR agent if clinically indicated, and an anti-VEGF agent). 2. Participants with a known MSI-H status must have received treatment with an immune checkpoint inhibitor (if locally indicated and available). 3. Eastern Cooperative Oncology Group Performance Status (ECOG PS) ≤ 1 . 4. Adequate hematologic, hepatic, and renal function.

4.2 Exclusion Criteria 1. History of malignancy within 3 years before enrollment (exceptions include cured basal/squamous cell skin carcinomas, in situ cervical carcinoma, or benign prostate neoplasm). 2. Known brain metastases, except those treated locally, clinically stable for at least 4 weeks prior to treatment, and lacking ongoing neurological symptoms. 3. Diarrhea (liquid stool) or ileus Grade > 1 . 4. Cerebrovascular accident/stroke < 6 months prior to enrollment.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: M9140 administered every 3 weeks (Q3W). The study will test 2 dose levels; if the low dose level is tolerated, escalation to the high dose level will occur. **Route of Administration:** Intravenous (IV) infusion **Duration of Treatment:** Until disease progression, unacceptable toxicity, withdrawal of consent, or any criterion for withdrawal from the study is met.

5.2 Concomitant & Prohibited Therapy Permitted: Standard supportive care for managing disease-related symptoms and ADC-associated toxicities. **Prohibited:** Other concurrent investigational agents or concurrent systemic anti-cancer therapies.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, Baseline Scans (Tumor Assessment), Labs
Baseline	Day 0	Enrollment confirmation, First Dose of M9140, Baseline PK sampling
Interim	Every 3 Weeks	M9140 IV administration, Safety Labs, PK/PD sampling, AE monitoring
End of Study	End of Treatment	Final Tumor Assessment, 30-Day Safety Follow-up, Survival Follow-up

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Assessment of safety and tolerability, determined by the incidence of Dose-Limiting Toxicities (DLTs) and Treatment-Emergent Adverse Events (TEAEs). **Measurement Method:** Laboratory evaluations, vital signs, and CTCAE clinical grading evaluated throughout the treatment period.

7.2 Secondary & Exploratory Endpoints * Pharmacokinetics (PK) profiling of total antibody and conjugated intact M9140. * Preliminary antitumor activity, including Objective Response Rate (ORR) and Progression-Free Survival (PFS) evaluated per RECIST v1.1 criteria.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Standard collection via clinical assessments at each Q3W visit. Special attention will be paid to potential hematological toxicities typical of TOP1 inhibitor payloads. **Serious Adverse Events (SAEs):** Must be reported within 24 hours of investigator awareness. **Data Monitoring Committee (DMC):** Internal safety review team to assess DLTs and make decisions regarding escalation from the low dose level to the high dose level.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Safety Set (all patients who receive at least one dose of M9140); Efficacy/ITT Set. **Sample Size Justification:** Approximately 12 patients. The sample size is not based on statistical power but is considered adequate for a Phase 1 open-label safety and dose-escalation/confirmation study in a regional subgroup. **Handling of Missing Data:** Standard oncology phase 1 conventions; generally, missing safety data will not be imputed.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Routine onsite and remote monitoring to verify protocol compliance, AE reporting, and the integrity of PK/efficacy data. **Audit Trail:** Continuous data management with eCRF locking, tracking of protocol deviations, and standardized data clarification forms.

11. Study Identifiers & Governance

Primary ID: PROCEADE-CRC-02 **Registry Number:** NCT06806046 **Sponsor/Lead Organization:** Merck KGaA, Darmstadt, Germany **Study Title:** Anti-CEACAM5 ADC Precentabart Tocentecan (M9140) in Chinese Participants With Solid Tumors **Official Regulatory Title:** A Phase 1, Open-label Study of Anti-CEACAM5 Antibody-Drug Conjugate Precentabart Tocentecan (M9140) in Chinese Participants With Solid Tumors (PROCEADE-CRC-02)

12. Clinical Framework (Oncology/Colorectal Cancer Specific)

Primary Condition: Locally Advanced or Metastatic Colorectal Cancer (CRC) **Diagnosis Threshold:** Histopathological confirmation of advanced CRC refractory to fluoropyrimidine, irinotecan, platinum agents, anti-EGFR, and anti-VEGF. **Medical Methodology:** Antibody-Drug Conjugate (ADC) targeted therapy. **Optimization Target:** Specific delivery of exatecan (TOP1 inhibitor) to CEACAM5-expressing tumor cells to disrupt DNA replication.

13. Study Procedures & Methodology

Management Pathway: Targeted systemic oncology pathway for refractory CRC (ADC administration Q3W). **Education Protocol (HCP):** Site initiation visits covering ADC preparation, infusion protocols, and early recognition of dose-limiting toxicities. **Education Protocol (Patient):** Rigorous informed consent procedure, side-effect management counseling, and AE self-reporting education. **Quality Audit Mechanism:** Independent medical monitoring, central PK parameter evaluation, and structured tumor imaging audits.

14. Observation & Follow-Up Schedule

Total Duration: Until disease progression or unacceptable toxicity, followed by survival follow-up. **Onsite Visit Cadence:** Every 3 weeks (Day 1 of each 21-day cycle). **Remote Monitoring Frequency:** Between treatment cycles as clinically indicated. **Checklist Review Interval:** Radiographic tumor assessments roughly every 6-8 weeks (standard oncology cadence) to determine continuing eligibility.

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	____	[DD-MMM-YYYY]				

Clinical Operations Lead | [Name] | ____ | [DD-MMM-YYYY] | | Lead
Statistician | [Name] | _____ | [DD-MMM-YYYY] |